KARTHIK THAKKALAPELLY

Generative AI Engineer | LLM & RAG Applications

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Professional Summary

Generative AI Engineer with hands-on experience designing and building end-to-end Retrieval-Augmented Generation (RAG) applications using Python, LangChain, ChromaDB, Hugging Face embeddings, and the Google Gemini API. Skilled in prompt engineering, semantic search, vector database design, and LLM orchestration for scalable, production-oriented AI applications. Azure Databricks-certified with direct exposure to implementing RAG pipelines on cloud AI platforms. Brings a strong analytical and data-engineering foundation to applied AI development, with a focus on building explainable, modular systems that solve real-world information-retrieval problems.

Technical Skills

Programming: Python

Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, LangChain Community, LangChain Chroma, Hugging Face Embeddings, Google Gemini API

Vector Databases: ChromaDB

AI Frameworks & Tools: Streamlit, PyPDF, Recursive Character Text Splitter

Libraries: Transformers, Sentence Transformers

Cloud & AI Platforms: Azure Databricks

Development Tools: Git, GitHub, VS Code

Generative Ai Project

Multi-Document RAG Assistant | LangChain, ChromaDB, Google Gemini API, Streamlit, Hugging Face, Python

Designed and developed an end-to-end Retrieval-Augmented Generation (RAG) application enabling natural language question-answering across multiple uploaded PDF documents, addressing inefficient information search and retrieval across large document sets.

Built a dynamic multi-PDF ingestion pipeline using PyPDF for text extraction and a Recursive Character Text Splitter for intelligent document chunking, improving retrieval accuracy and search efficiency.

Generated document embeddings with Hugging Face Sentence Transformers and indexed them in a ChromaDB vector database, enabling fast semantic similarity search for multi-document retrieval.

Integrated the Google Gemini API through LangChain orchestration, applying prompt engineering techniques to produce accurate, context-grounded answers from retrieved passages.

Developed an interactive Streamlit chat interface with source page citations, confidence score calculation, and a retrieved-context viewer, improving transparency and answer traceability.

Architected a modular, session-state-managed system with dynamic vector database generation and robust error handling, supporting scalability across concurrent multi-document sessions.

Designed the codebase for extensibility, laying groundwork for planned enhancements: hybrid search, OCR for scanned PDFs, metadata filtering, persistent chat memory, cloud deployment, and authentication.

Certifications

Implement Retrieval Augmented Generation (RAG) with Azure Databricks — Microsoft Learn, July 2026

Explore Azure Databricks — Microsoft Learn, July 2026

Micro Prompt Engineering — FreeAcademy.ai, June 2026

Data Science and Artificial Intelligence Certification — AI LINC, February 2026

Education

Bachelor of Technology (B.Tech) — Information Technology 2022 – 2026

KU College of Engineering and Technology, Kakatiya University, Hanamkonda | CGPA: 9.08 / 10.0